

CHAPTER 9

COUNTING AND PROBABILITY

9.8

Probability Axioms and Expected Value

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Probability Axioms

Let S be a sample space. A **probability function** P from the set of all events in S to the set of real numbers satisfies the following three axioms: For all events A and B in S :

1. $0 \leq P(A) \leq 1$.
2. $P(\emptyset) = 0$ and $P(S) = 1$.
3. If A and B are disjoint (that is, if $A \cap B = \emptyset$), then the probability of the union of A and B is

$$P(A \cup B) = P(A) + P(B).$$

Example 9.8.2 – The Probability of the Complement of an Event

Suppose that A is an event in a sample space S . Deduce that $P(A^c) = 1 - P(A)$.

Example 9.8.2 – Solution

S representing the universal set U ,

$$A \cap A^c = \emptyset \quad \text{and} \quad A \cup A^c = S.$$

Thus S is the disjoint union of A and A^c , and so

$$P(A \cup A^c) = P(A) + P(A^c) = P(S) = 1.$$

So $P(A) + P(A^c) = 1$ and subtracting $P(A)$ from both sides gives the result that $P(A^c) = 1 - P(A)$.

Probability Axioms and Expected Value

Probability of the Complement of an Event

If A is any event in a sample space S , then

$$P(A^c) = 1 - P(A).$$

9.8.1

Probability of a General Union of Two Events

If S is any sample space and A and B are any events in S , then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

9.8.2

Suppose a card is chosen at random from an ordinary 52-card deck. What is the probability that the card is a face card (jack, queen, or king) or is from one of the red suits (hearts or diamonds)?

Example 9.8.4 – *Solution*

Let A be the event that the chosen card is a face card, and let B be the event that the chosen card is from one of the red suits. The event that the card is a face card or is from one of the red suits is $A \cup B$.

Now $N(A) = 4 \cdot 3 = 12$ (because each of the four suits has three face cards), and so $P(A) = 12/52$.

Also, $N(B) = 26$ (because half the cards are red), and so $P(B) = 26/52$.

Example 9.8.4 – Solution

continued

Finally, $N(A \cap B) = 6$ (because there are three face cards in hearts and another three in diamonds), and so $P(A \cap B) = 6/52$. It follows from the formula for the probability of a union of any two events that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{12}{52} + \frac{26}{52} - \frac{6}{52} = \frac{32}{52} \cong 61.5\%.$$

Thus the probability that the chosen card is a face card or is from one of the red suits is approximately 61.5%.



Expected Value

Definition

Suppose the possible outcomes of an experiment, or random process, are real numbers $a_1, a_2, a_3, \dots, a_n$, which occur with probabilities $p_1, p_2, p_3, \dots, p_n$. The **expected value** of the process is

$$\sum_{k=1}^n a_k p_k = a_1 p_1 + a_2 p_2 + a_3 p_3 + \cdots + a_n p_n.$$

Example 9.8.5 – *Expected Value of a Lottery*

Suppose that 500,000 people pay \$5 each to play a lottery game with the following prizes: one grand prize of \$1,000,000, 10 second prizes of \$1,000 each, 1,000 third prizes of \$500 each, and 10,000 fourth prizes of \$10 each.

What is the expected value of a ticket?

Example 9.8.5 – Solution

Each of the 500,000 lottery tickets has the same chance as any other of containing a winning lottery number, and so $p_k = \frac{1}{500,000}$ for each $k = 1, 2, 3, \dots, 500,000$. Let a_i be the net gain for an individual ticket a_i , where $a_1 = 999,995$ (the net gain for the grand-prize ticket, which is one million dollars minus the \$5 cost of the winning ticket), $a_2 = a_3 = \dots = a_{11} = 995$ (the net gain for each of the 10 second-prize tickets), $a_{12} = a_{13} = \dots = a_{1,011} = 495$ (the net gain for each of the 1,000 third-prize tickets), and $a_{1,012} = a_{1,013} = \dots = a_{11,011} = 5$ (the net gain for each of the 10,000 fourth-prize tickets).

Example 9.8.5 – Solution

continued

Since the remaining 488,989 tickets just lose \$5, $a_{11,012} = a_{11,013} = \dots = a_{500,000} = -5$. The expected value of a ticket is therefore

$$\sum_{k=1}^{500,000} a_k p_k = \sum_{k=1}^{500,000} \left(a_k \cdot \frac{1}{500,000} \right) \quad \text{because each } p_k = 1/500,000$$

$$= \frac{1}{500,000} \sum_{k=1}^{500,000} a_k \quad \text{by theorem 5.1.1(2)}$$

Example 9.8.5 – Solution

continued

$$\begin{aligned} &= \frac{1}{500,000}(999,995 + 10 \cdot 995 + 1,000 \cdot 495 + 10,000 \cdot 5 + (-5) \cdot 488,989) \\ &= \frac{1}{500,000}(999,995 + 9,950 + 495,000 + 50,000 - 2,444,945) \\ &= -1.78. \end{aligned}$$

In other words, a person who continues to play this lottery for a very long time will probably win some money occasionally but on average will lose \$1.78 per ticket.